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Validation of Analytical Methods

Validation of analytical methods is a subject of considerable interest. Documents such as the "ACS Guidelines for Data Acquisition and Data Quality Evaluation" (1) recommend the use of validated methods. The promulgation of federal environmental regulations requires the inclusion of validated reference methods. Standards-writing organizations spend considerable time in collaborative testing of methods they prepare, validating them in typical applications and determining their performance characteristics. Nevertheless, questions about the appropriateness of methods and the validity of their use in specific situations often arise. Some of these questions may be due to differences in understanding both what a method really is and what the significance of the validation process is. This paper attempts to clarify the nomenclature of analytical methodology and to define the process of validating methods for use in specific situations.

Hierarchy of Methodology

The hierarchy of methodology, proceeding from the general to the specific, may be considered as follows: *technique* → *method* → *procedure* → *protocol*.

A **technique** is a scientific principle that has been found to be useful for providing compositional information; spectrophotometry is an example. Analytical chemists historically have investigated new measurement techniques for their ability to provide novel measurement capability, or to replace or supplement existing methodology. As a result of innovative applications, analysts can now analyze

for myriad substances in exceedingly complex mixtures at ever lower trace levels, with precision and accuracy undreamed of only a few years ago (2).

A **method** is a distinct adaptation of a technique for a selected measurement purpose. The pararosanine method for measurement of sulfur dioxide is an example. It involves measuring the intensity of a specific dye, the color of which is "bleached" by the gas. Several procedures for carrying out this method may be found in the literature. Modern methodology is

complex and may incorporate several measurement techniques; a method may thus be interdisciplinary.

A **procedure** consists of the written directions necessary to utilize a method. The "standard methods" developed by ASTM and AOAC are, in reality, standardized procedures. ASTM D2914—Standard Test Method for the Sulfur Dioxide Content of the Atmosphere (West-Gaeke Method)—is an example (3). While a precise description is the aim, it is difficult, if not impossible, to describe every de-

Hierarchy of Analytical Methodology		
	Definition	Example
Technique	Scientific principle useful for providing compositional information	Spectrophotometry
Method	Distinct adaptation of a technique for a selected measurement purpose	Pararosanine method for measurement of sulfur dioxide
Procedure	Written directions necessary to use a method	ASTM D2914—Standard Test Method for the Sulfur Dioxide Content of the Atmosphere (West-Gaeke Method)
Protocol	Set of definitive directions that must be followed, without exception, if the analytical results are to be accepted for a given purpose	EPA Reference Method for the Determination of Sulfur Dioxide in the Atmosphere (Pararosanine Method)

This REPORT is based on a talk given at the 184th ACS National Meeting, Sept. 12–17, 1982, Kansas City, Mo.

tail of every operation in a procedure. Accordingly, some level of sophistication is presumed for the user of every published procedure; if very sophisticated users are contemplated, only a minimum of detail will be provided and vice versa. However, it should be noted that any omission in the description of critical steps is a potential source of variance or bias, even in the hands of knowledgeable analysts. Because of the flexibility intentionally or unintentionally provided to the analyst, or because of differences in interpretation, it is fair to say that minor-to-major differences of application occur in the use of even the most precisely defined procedures. Such differences often account for the interlaboratory variability observed in many collaborative tests. Further, at some point of departure from a published procedure, a new method results that may need its own validation.

The term **protocol** is the most specific name for a method. A protocol is a set of definitive directions that must be followed, without exception, if the analytical results are to be accepted for a given purpose. Protocols may consist of existing methods or procedures, modifications of such, or they may be developed especially for specific purposes. Typically, they are prescribed by an official body for use in a given situation such as a regulatory process. The EPA Reference Method for the Determination of Sulfur Dioxide in the Atmosphere (Pararosaniline Method) is an example of a protocol (4). The test method specified as part of a contractual arrangement for the acceptance of data or a product or material is another example of a protocol, although it may not be called that in the contract.

A plethora of methods, procedures,

and protocols based on the same measurement principle can arise for a given analytical determination. Usually, they are worded differently, and they may contain subtle or major differences in technical details. The extent to which each needs to be individually validated is a matter of professional judgment. It is evident that some validation tests could be merely a matter of experimentally testing the clarity of the written word.

Goals for Validation

Validation is the process of determining the suitability of methodology for providing useful analytical data. This is a value judgment in which the performance parameters of the method are compared with the requirements for the analytical data, as illus-

trated in Figure 1. Obviously a method that is valid in one situation could be invalid in another. Accordingly, the establishment of firm requirements for the data is a prerequisite for method selection and validation. When data requirements are ill-considered, analytical measurement can be unnecessarily expensive if the method chosen is more accurate than required, inadequate if the method is less accurate than required, or utterly futile if the accuracy of the method is unknown.

Fortunately, typical and even standard measurement problems often exist. Examples include a wide variety of clinical analyses, environmental determinations, and recurring measurements for the characterization of industrial products. The kinds of samples for which methods have been val-

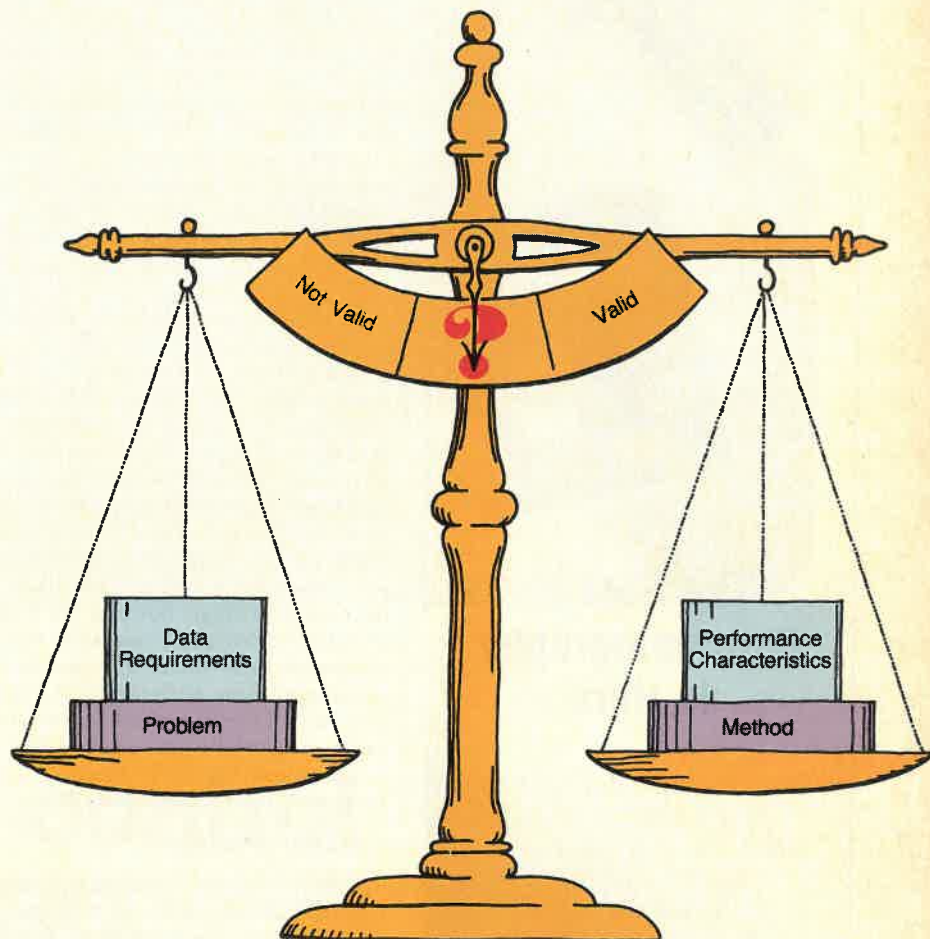


Figure 1. Basic concept of the validation process

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idated should be clearly described, and users should be aware of the need to demonstrate their own abilities to use the method in their own laboratories.

Statements of precision and accuracy are often a result of a validation process, especially in the case of a collaborative test exercise. Such statements are often misinterpreted; they merely describe the results of the exercise and are, at best, estimates of typical performance expectations for the method. They should not be construed to be performance parameters nor should they be used to estimate the uncertainty of any future data obtained by using the method. However, information on precision and accuracy should be obtained to the extent possible since it provides a quantitative basis for judging general performance capability.

Other information useful for characterizing methodology or for judging its suitability for a given use includes: sensitivity to interferences, limits of detection, and useful range of measurement. The specific details for evaluating methodology in these respects are beyond the scope of the present paper. Ordinarily, such information is best obtained as a result of applied research during the method development stage. Because the limit of detection is closely related to the attainable precision at the lower limit of measurement, both the limit of detection and the lowest concentration range measurable (often called limit of quantitation) should be evaluated, as pertinent, in every laboratory (1, 5).

Validation Process

The validation process verifies that the methodology is based on sound technical principles and that it has been reduced to practice for practical measurement purposes. Both the need to validate methodology and the procedure to be followed are matters for professional judgment. The validation can be either general or specific.

General Validation. Validation of measurement techniques depends on the elucidation of the scientific principles upon which they are based. Such validation results from the research of the scientific community, and its soundness is evaluated by peer review. Better understanding of measurement principles can extend their scope and improve the quality of their use. To confirm the above statement, one need only think about the varied research that has contributed to the understanding of the principles of gas chromatography and that has led to development of its status as a prime measurement technique.

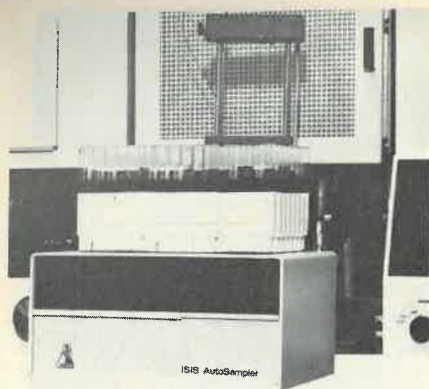
Methods arise as the result of applied research, typically by individuals, that often involves both a compre-

hensive understanding of measurement techniques and a high degree of ingenuity and innovation in their application. Testing of the methods in typical practical situations plays a key role in both the development process and in validation. While ordinarily limited in scope, validation at the research stage can be comprehensive and can apply to a wide variety of end uses.

Procedures are developed for the end use of methods in practical analytical situations. The user laboratory ordinarily needs more experimental details than are contained in a published research report of a method to use it in practical measurements. Frequently, as a method gains widespread use, procedures evolve that the users may decide need to be standardized. This is often done by consensus in a standards organization forum. During this process, the resulting standard procedure is examined both technically and editorially. A thorough review process includes collaborative testing in which typical stable test materials are analyzed to verify the procedure's usefulness and to identify both technical and editorial weaknesses. The process is illustrated in Figure 2. If the composition of the reference samples is known, precision and bias, both intra- and interlaboratory, can be evaluated; otherwise, only precision can be evaluated. If a method of known accuracy is available, the collaborative test may consist of its comparison with the candidate method, in which case both precision and bias can be evaluated. The performance parameters of the procedure so evaluated are for the conditions of the collaborative test that are considered typical. Any extension of them to other kinds of samples is by inference only, and may need to be justified. Although it can be time-consuming, the development of a standard method is one of the best ways to validate a procedure because of the breadth of examination that is involved.

A protocol is prescribed by fiat of an organization requiring a specific kind of measurement. Presumably it results from an intelligent decision based on the organization's validation process or that of others. This may consist of an extensive collaborative test or publication of a proposed protocol for public comment. Unfortunately, expediency has overruled sound scientific judgment in some cases, resulting in the promulgation of unvalidated and scientifically defective protocols (6). Protocols that are specified in a contractual arrangement may be selected arbitrarily or through a well-conceived selection process. Verification of their validity for the specific use should be a prime consideration.

Validation for Specific Use. The



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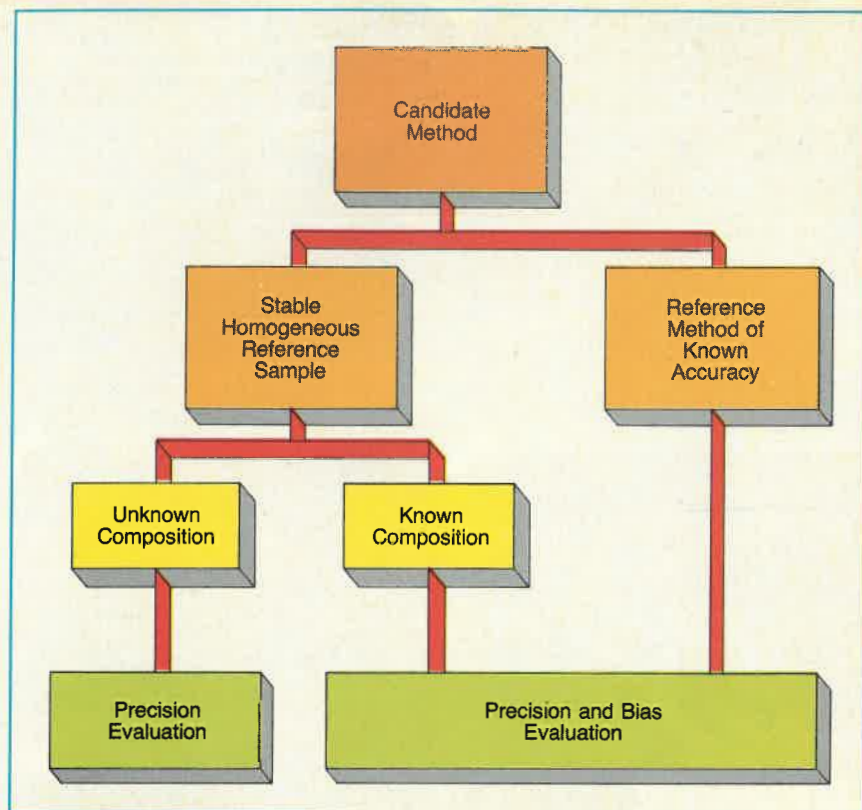


Figure 2. Collaborative test process

ultimate use of analytical methodology is to produce compositional information about specific samples necessary for the solution of particular problems ranging from exotic research investigations to the very mundane. The selection of appropriate measurement methodology is often a major consideration. Methods or procedures, even if previously validated in general terms, cannot unequivocally be assumed to be valid for the situation in hand, because of possible differences in sample matrix and other considerations. Professional analytical chemists traditionally have recognized this and their responsibility to confirm or prove (if necessary) both the validity of the methodology used for specific application (2) and their own ability to reduce it to practice.

The classical validation process is illustrated in Figure 3. When reference samples are available that are similar in all respects to the test samples, the process is very simple: It consists of analyzing a sufficient number of reference samples and comparing the results to the expected or certified values (7). Before or during such an exercise, the analyst must demonstrate the attainment of a state of statistical control of the measurement system (8) so that the results can be relied upon as representative of those expected when using the methodology-measurement system.

When a suitable reference material is not available, several other approaches are possible. One consists of comparing the results of the candidate method with those of another method known to be applicable and reliable, but not useful in the present situation because of cost, unavailability of personnel or equipment, or other reasons. Even the agreement of results with those obtained using any additional independent method can provide some useful information.

Spiked samples and surrogates may be used as reference samples. This approach is less desirable and less satisfactory because of the difficulty in the reliable preparation of such samples and because artificially added materials such as spikes and surrogates may exhibit matrix effects differing from those of natural samples. Split samples of the actual test samples may be used to evaluate the precision of a method or procedure, but they provide no information about the presence or magnitude of any measurement bias.

Another approach is to infer the appropriateness of methodology from measurements on analogous but dissimilar reference materials. The critical professional judgment of the analyst is necessary to decide the validity of the inference.

In all cases, sufficient tests must be made to evaluate the methodology for

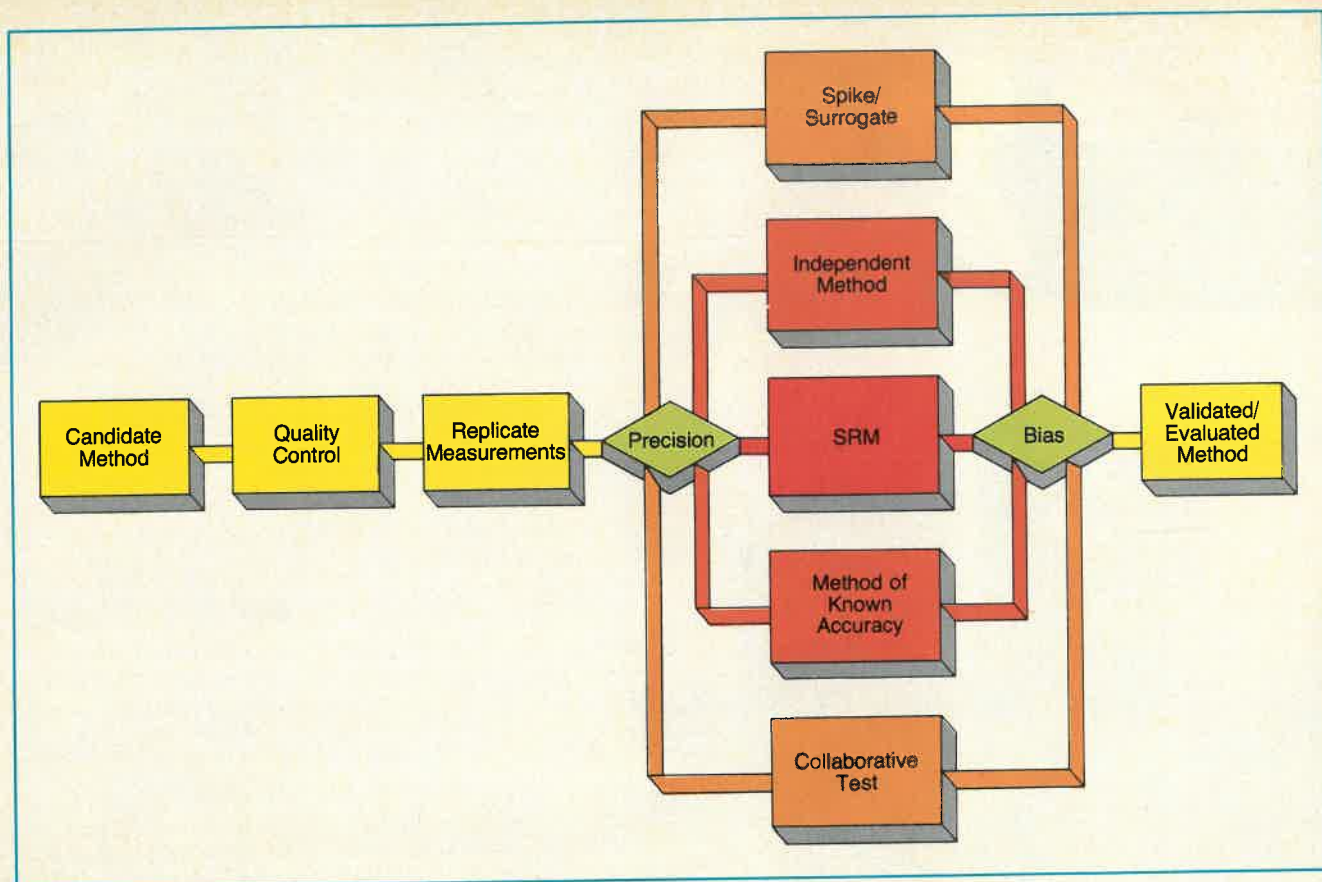


Figure 3. General process for evaluation/validation of methodology

the variety of matrices and ranges of composition expected during the measurement process. Ordinarily, the latter should include three levels of concentration, namely, the extremes and the mid-range of compositions expected. Statistical considerations suggest that at least six degrees of freedom (ordinarily seven measurements) should be involved at each decision point.

Conclusion

A valid method is necessary but not sufficient for the production of valid data. Most methods require a degree of skill on the part of the analyst; this skill constitutes a critical factor in the measurement process. It is common knowledge that data obtained by several laboratories on the same test sample using the same methodology may show a high degree of variability. The alleviation of such a problem is in the area of quality assurance of the measurements (8). Data obtained by a valid method used in a well-designed quality assurance program should allow the assignment of limits of uncertainty that can be used to judge the data's validity.

It should be remembered that the validity of any data will also depend upon the validity of the model and of the sample (8, 9). The model represents the conceptualization of the

problem to be solved, describes the samples that should be analyzed, the data base required, and the way the model will be utilized. Obviously, even flawless measurement data will be of little value if the basic concepts are faulty. Likewise the samples analyzed must be valid if the results obtained for them are to be intelligently interpreted.

The key role of reliable reference materials in the validation of analytical measurements cannot be overemphasized. Their use in validating the methodology has already been discussed. A planned sequential analysis of reference materials in a quality assurance program can assess the quality of the data output and thus validate the overall aspects of the analytical measurement system (7).

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